

Homework 1 - Directions: Look in the top right of this document. That is your due date and total points available. Turn it in no later than midnight of that date. To turn in your assignment you should upload your document to the ASULearn dropbox this file is attached to. To do so, you may take a picture, scan your work, or otherwise convert your work to a .pdf, .jpg, or other common image file. A .pdf is preferred.

You may either...

...print this sheet out and do your work directly on it and upload the completed document.

...do all work on a separate sheet of paper and upload your work when completed.

...type your work using an editor capable of displaying the symbols needed to communicate your work to me or use a digital workspace to edit this file directly. (Not for the faint of heart.)

You may use all resources available to you. Unless otherwise stated I expect your work to be shown. Calculus II is about the journey, not the destination.

1) [No work required] (5 points.) Office hours are a thing that many university students don't take enough advantage of. I can only imagine it's because of the connotations of 'being sent to the office' in high school and how that leaves an icky feeling with students. But office hours are meant to be a place where we can discuss troublesome ideas or problems. Currently, my office hours are listed on the course policy file on ASULearn. Please answer the following two questions.

a) Do the times posted give you some ability to communicate with me?

Yes

b) If not, please make a note of a time frame you would be available in. I can usually carve out times with 24 hours notice but want to have a general idea of when not to schedule other appointments.

2) Use the product rule to find the derivative of $f(x) = xe^x$. (5 points.)

$$f'(x) = \frac{d}{dx} (xe^x)$$

$$f'(x) = \frac{d}{dx} (x)(xe^x) + x \cdot \frac{d}{dx} (e^x)$$

$$f'(x) = 1e^x + xe^x$$

$$f'(x) = e^x + xe^x$$

3) Use the chain rule to find the derivative of $f(x) = \tan(e^x + x^2)$. (5 points.)

$$f'(x) = \frac{d}{dx} (\tan(e^x + x^2))$$

$$f'(x) = \frac{d}{dy} (\tan(y)) \cdot \frac{d}{dx} (e^x + x^2)$$

$$\tan^2 \theta + 1 = \sec^2 \theta$$

$$f'(x) = \sec(y)^2 \cdot \frac{d}{dx} (e^x + x^2)$$

$$f'(x) = \sec(y)^2 \cdot (e^x + 2x^1) \quad \text{power rule}$$

$$f'(x) = \sec(e^x + x^2)^2 \cdot (e^x + 2x)$$

4) Use the Fundamental Theorem of Calculus to evaluate the following. (5 points.)

$$\frac{d}{dx} \int_4^{x^2} \sin(t) dt$$

5) Evaluate the following definite integral. (5 points.)

$$\int_{-3}^9 dx$$

It's late on Sunday night, I didn't leave enough time for this, and I refuse to use AI for this class so I will just say that that's all I got this week. (work)

6) Complete problem 74 on page 37. Be sure to read the directions above the batch of problems. (5 points.)

7) Complete problem 82 on page 37. Hint: Read the directions and plot that graph in desmos. (5 points.)

8) Simplify the following by using a mixture of the properties of definite integrals. (5 points.)

$$\int_3^8 f(x) \, dx + \int_{-1}^3 f(x) \, dx + \int_8^{-1} f(x) \, dx$$

9) Complete problem 90 on page 37. Be sure to read the information in the blue text above the problem. (5 points.)

$$\int_0^2 (f(x) - g(x)) \, dx$$

10) Complete problem 93 on page 37. Be sure to read the information in the blue text above the problem. (5 points.)

$$\int_2^4 (4f(x) - 3g(x)) \, dx$$